Heliyon 10 (2024) e31053

Contents lists available at ScienceDirect

Heliyon

journal homepage: www.cell.com/heliyon

Review article

Investigating the integration of artificial intelligence in English as foreign language classes for enhancing learners' affective factors: A systematic review

Lujain AlTwijri *, Talal Musaed Alghizzi

Imam Mohammad Ibn Saud Islamic University, Riyadh, Saudi Arabia

ARTICLE INFO

Keywords:
Artificial intelligence
English as a foreign language
Affective factors
Higher education

ABSTRACT

Integrating Artificial Intelligence (AI) applications into language learning and teaching is currently a growing trend in higher education. Literature reviews have demonstrated the effectiveness of AI applications in improving English as a foreign language (EFL) and English as a second language (ESL) learners' receptive and productive skills, vocabulary knowledge, and intercultural competencies. However, systematic reviews investigating the usefulness of AI technologies in higher education to enhance EFL learners' affective factors are scarce. This study is a systematic review that investigates the effectiveness of integrating AI technologies to enhance EFL learners' motivation, engagement, and attitude, and reduce their learning anxiety. Articles from reputable journal databases such as IEEE, Wiley, Web of Science, Sage, ProQuest, Springer, and Science Direct were screened by examining titles and abstracts, and irrelevant articles were excluded from the search. Of the 64 articles analyzed only 21 articles published between 2017 and 2023 were determined to be relevant to the research topic. The findings suggest that the implementation of AI technologies in EFL contexts is in its early stages, and further research is required to establish the impact of AI-integrated classes on EFL learners' affective factors. This review also identifies the gaps in literature and recommends avenues for future research in this novel area.

1. Introduction

Learners of EFL choose to learn English for various reasons, including personal interest, social needs, career objectives, or academic obligations. Although EFL teachers work extremely hard to assist them, some students struggle and ultimately fail in their classes [1]. According to the literature, success in learning a second language (L2) is determined by numerous factors like environmental, cognitive social and affective factors [2]. Factors such as motivation, engagement, attitudes, anxiety, and self-esteem are referred to as affective factors as "the way we feel about our capacities and ourselves can either facilitate or impede our learning" (2, p. 8). Among the most promising technologies that can be integrated to foster EFL learners' affective factors in higher education is Artificial Intelligence (AI). AI has the potential to customize language learning and meet different learners' needs by creating optimal learning opportunities [3].

* Corresponding author.
E-mail addresses: altwijri-lujain@hotmail.com (L. AlTwijri), tmalghizzi@imamu.edu.sa (T.M. Alghizzi).

https://doi.org/10.1016/j.heliyon.2024.e31053
Received 11 December 2023; Received in revised form 5 May 2024; Accepted 9 May 2024
Available online 10 May 2024
2405-8440/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1.1. Affective factors

The affective factors play a significant role in language learning as they interplay between the learner's feelings, thinking, and acting [3], and "there is no separation of mind and emotions; emotions, thinking and learning are all linked" (4, p.71). Therefore, connecting learning with the affective factors will facilitate language learning in different ways. First, it is neurobiologically believed that human brain rarely, if ever, remembers things that do not have emotional connections because without emotions, there will be no curiosity, awareness, and attention, which are all critical components in the learning processes [3]. In addition, to get the learner's attention to the necessary learning input, the human brain needs to be connected to meaningful experiences to "activate the brain and facilitate recall" (3, p.6) and here comes the role of the affective factors that engage with meaning and make every learning experiences meaningful.

Acknowledging the importance of affective factors has led to the realization that it is crucial to not only concentrate on the outcomes of learning but also on the unique needs of learners and their search for meaning. Therefore, in addition to enhancing their methods for teaching language, teachers need to use a variety of techniques and teaching strategies that take a holistic approach to educating an individual. Teachers should take into account the affective needs of their students to assist them living prosperous lives and contribute to their society [1]. This systematic review considers four affective factors: motivation, engagement, attitude, and anxiety. The main reason for focusing on these four factors is that studies investigating the impact of AI applications on EFL learners in higher education have extensively pursued motivation, engagement, attitude, and anxiety. This could be due to the complexity of measuring other factors, which requires sophisticated research methods and tools that challenge implementation, or the lack of interdisciplinary collaboration.

1.1.1. Motivation

Motivation refers to the reason underlying willing and volitional behavior [4]. Motivation is the driving force that steers and shapes individuals' actions towards achieving specific objectives. It energizes them to engage in work, strive for success, initiate movement, set goals and steer towards them, and persist with efforts. When a learner is motivated, this manifests through his/her commitment and in his/her behavioral, emotional and cognitive involvement [5]. Motivation plays a crucial role in varying success levels in language learning, as motivated individuals learn more efficiently and sustain their learning activities over time. Such motivated learners are characterized by optimism, persistence, cooperation, overcoming obstacles, and keenness to acquire new knowledge [6]. According to [7], motivation is the willingness to learn a second language and the diligence applied to continue this learning.

According to [8], self-determination theory is a theoretical approach to motivation and personality in humans that emphasizes the significance of individuals' "evolved inner resources for personality development and behavioral self-regulation" (p.68). The self-determination theory distinguishes between two forms of motivation: intrinsic motivation and extrinsic motivation. Intrinsic motivation is considered the most effective and stems from internal fulfillment and pleasure. It is linked with higher levels of achievement as it originates from personal interest and self-derived satisfaction [9]. In contrast, extrinsic motivation guides human actions toward set goals to obtain rewards or meet deadlines. Individuals who are extrinsically motivated have less autonomy and pursue language learning for external reasons like seeking job promotions, gaining acknowledgment and applause, meeting the expectations of others, or the need for social inclusion and connection [9].

1.1.2. Engagement

Engagement pertains to the quantity and quality of learners' active involvement in a language learning experiences [10]. It refers to the degree of attention and interest that learners show towards their learning activities and tasks. It is usually associated with high levels of cognitive processing and retention of information [10]. In addition, meaningful learning is unlikely to occur without the active participation and commitment of engaged learners. Engaged learners are more likely to actively participate in class, explore topics in depth, and seek learning opportunities. Consequently, understanding and promoting engagement is a major focus in educational research and practice, aiming to improve language learning outcomes.

1.1.3. Attitude

Attitude is a cognitive and affective factor that is dimensional. Attitudes vary in degree of favorability and predispose individuals to act in a specific way. It is learned rather than inherited and can be modified through experience [11]. [12] claims that EFL learners' attitudes manifest toward a variety of things, such as the target language, the speakers and the culture of the target language, and the social significance of learning the L2.

1.1.4. Anxiety

Anxiety is viewed as a feeling of uneasiness, worry, and apprehension that is affected by intrinsic and extrinsic factors related to the foreign language student. Students may experience this fear or worry when performing in an L2 and may feel uncomfortable. Such feelings were identified in three performances in foreign language learning: communication apprehension, when learners were unable to express ideas and thoughts; test anxiety; and apprehension of negative evaluation arising from the need to leave a positive social impression [11].

1.2. Artificial intelligence

Artificial intelligence (AI) is a field of study that focuses on addressing cognitive challenges similar to those solved by human

intelligence to create programs that mimic human thought processes. It aims to understand the nature of human intelligence, including its different structures and aspects [13]. AI seeks to replicate human cognitive abilities in practical scenarios, enabling the simulation of specific skills like speech recognition. These cognitive capabilities are then converted into computational formulas to solve problems and make decisions. The short-term goal of this field of study includes the creation of intelligent-based applications, while its long-term goal is to achieve a human-level of intelligence [14].

1.3. The features of AI

The features of AI are strongly linked to its implementation in education; it provides a range of benefits and transforms traditional learning environments into more effective, individualized, and accessible settings. Based on the most recent research in this field, there are different areas where AI applications can improve learning and teaching activities [15].

First, AI analyzes various amounts of data to identify patterns and make judgments. This feature allows for the personalization of learning experiences. After evaluating each student's performance, preferences, and learning style, AI systems can customize the learning path, content, and pacing to suit their particular needs. Second, AI efficiently automates repetitive tasks, such as grading, to direct educators' focus on more complex tasks. Another distinctive feature of AI is adapting learning platforms based on the learner's level [15]. This feature promotes efficient learning by adjusting the tasks' difficulty level, thus providing challenges ideally suited to the learner's current skill level. In addition, AI can comprehend and generate human language to facilitate interactions and provide prompt feedback. Furthermore, AI makes educational materials more accessible by converting text to speech and speech to text and recognizing visual materials. Lastly, AI is very effective in helping educators understand learners' performance and identify trends and insights to adjust the teaching process accordingly [16].

1.4. Artificial intelligence in higher education

The advancement of technologies and the increased processing capacity of machines significantly impact higher education. AI enhances the higher education sector by motivating educators and students to engage profoundly [17], and presenting opportunities for the teaching and learning processes. AI holds the promise of enhancing students' academic performance by providing personalized learning opportunities and identifying areas needing improvement [14].

In higher education, AI can significantly improve learning outcomes across various fields of study. It supports educators in providing guidance and tailor instructions to meet the diverse needs of learners. In addition, AI can analyze data and ensure that the teaching methods align with learner's unique learning styles and preferences. Therefore, it enhances learner engagement and fosters a deeper understanding of different learning materials [14]. Furthermore, AI applications play a significant role in supporting L2 learners by bridging gaps between their existing and new knowledge to adjust knowledge to the learner's current understanding or facilitate the acquisition of new knowledge. Moreover, AI-based learning environments encourage L2 learners to achieve higher proficiency levels, thus accelerating the learning process and decreasing teachers' workload [18].

1.5. Artificial intelligence in language learning

Various AI tools have been utilized in learning foreign languages [19], including automated writing-evaluation system, machine translation tool, intelligent learning management system, chatBot, and virtual learning environment [20]. However, very few of them have been specifically designed to create interactive activities to enhance learners' motivation, engagement, and attitude and reduce anxiety toward language learning.

AI-powered chatbot is among the most prevalent AI applications used to improve the affective factors of learning. It is a computer program equipped with AI capabilities to mimic human-like intelligence through text or voice. It provides an interaction experience to enrich users' knowledge. Numerous studies have shown that chatbots have been instrumental in developing EFL learners in different aspects of language domains such as grammar, vocabulary, and speaking skills ([21–24]). Furthermore, AI chatbots fostered learners' motivation, engagement, and positive attitudes toward learning EFL ([25–28]).

1.6. Purpose of the study

Success in learning EFL depends on many variables. These variables include external factors, such as the learning environment, teaching methods and techniques, and the learning materials. Apart from the aforementioned external factors, certain internal factors are specific to the learners and can exert positive or negative influences on the learning process of EFL. These include affective factors, such as motivation, aptitude, engagement, attitude, and anxiety. They are becoming more significant for learners and instructors as research into language acquisition and foreign language learning continues [29].

In addition, as AI applications have become integrated into the educational sector and their potential benefits for language learning are increasingly recognized, more studies have likely focused on investigating their impact on various learner's affective factors, such as motivation, attitude, engagement and anxiety [30]. It has been found that the integration of AI enhances affective factors and this attributed to different reasons, such as personalized and constant feedback, an interactive learning environment, accessibility and flexibility in terms of place and time, and autonomous learning opportunities. All this, in turn, fosters a positive attitude toward language learning, reduces anxiety, and enhances motivation and engagement [31]. The focus on these affective factors reflects a clear understanding of their significant role in EFL learning success. With the development of AI applications and continuing integration of

AI applications into higher education, studies in this area are crucial for developing learner-centered language learning environments.

AI applications have been developed to create opportunities for foreign-language learners to enhance their academic performance [17] and foster affective factors in learning EFL. Although there are significant advantages to AI applications, systematic reviews investigating their effectiveness in promoting EFL learners' affective factors are scarce. Furthermore, little is known about how AI technology is implemented in higher educational systems, how well L2 learners engage with AI technology, and how their engagement with AI fosters their motivation and attitude toward learning the language. Therefore, this study aims to investigate the integration of AI applications in higher education to enhance EFL learners' affective factors—such as motivation, engagement, and attitude—and reduce their learning anxiety.

To the best of our knowledge, only a few studies have reviewed the integration of AI into education ([16,32–36]). These studies provided new insights into how AI can positively transform teaching and learning processes, thus guiding future researchers, policy-makers, and educational practitioners. In addition, these studies focused on innovative AI applications, highlighting their significance in supporting personalized learning and assessment in different ways. Nevertheless, the findings are not generalizable due to the small sample size. Thus, it is challenging to draw conclusions about the effectiveness of AI at all levels of education. Furthermore, investigating the integration of AI into education poses significant challenges and ethical considerations that need to be assessed in long-term or longitudinal studies. Despite these weaknesses, the ongoing interest in AI's potential for education suggests that research in this area to continue to grow. Some researchers have examined the use of AI dialogue systems to enhance EFL learners' interactional competence [19]. Others have analyzed the opportunities and challenges of integrating AI into education ([32,34]). Notably, some systematic reviews emphasize the need for more research to understand the effectiveness of AI technologies in fostering learners' socio-affective factors, especially in higher education because higher education settings are not only about promoting cognitive skills and knowledge but also play a significant role in developing learners' socio-affective skills, which are crucial for their overall well-being ([35,37]). Furthermore, studies analyzing the effectiveness of AI technology in fostering learners' affective factors are scarce.

1.7. Research questions

This study aimed to identify the gaps in existing literature and to contribute to the understanding of the role of AI in enhancing EFL learners' affective factors. Thus, the following research questions guided this systematic review.

1 What are the roles of AI applications integrated into EFL classes?
2 To what extent are AI applications effective in enhancing EFL learners' motivation, attitude, and learning engagement?
3 To what extent are AI applications effectively reducing EFL learners' anxiety?

2. Methods

According to [38], a systematic review is a method for synthesizing knowledge that establishes exploratory research questions through deliberate selection, identification, and analysis of available data. A systematic review helps map the published literature, pinpoint research gaps, outline research goals, and offer recommendations [19]. To identify AI tools that can be integrated into EFL classes to enhance learners' motivation, engagement, and attitude and reduce their learning anxiety, this study employs the five-step method of [39], which is based on the grounded theory that allows for a comprehensive search and analysis of the articles collected from databases. The steps in this method are to identify the scope of the systematic review, conduct a search after selecting the final samples, evaluate the selected samples, and present the results [19].

2.1. Identifying the scope of the systematic review

The first step is to clarify the inclusion and exclusion criteria to identify relevant articles in the literature [19]. For inclusion, journal articles needed to be empirical, published in English, integrated with an AI application into EFL classes, applicable in the higher education sector, focused on learners' affective factors, and published recently owing to the rapid evolution of AI. The reason for investigating the effectiveness of AI technologies, which are used to enhance learners' motivation, attitude, and learning engagement and reduce anxiety, is that they significantly improve learners' performance and language proficiency when used in higher education.

On the other hand, articles that were not considered were published in other languages; did not integrate any of the AI applications; were applied in K–12 education; focused on developing the four basic skills, grammar, vocabulary; or were editorials or opinion articles. Editorials and opinion articles were excluded as they present the author's subjective perspective and lack methodological rigor [19]. Besides, [40] believes that including these potentially biased studies may affect the findings of the systematic review.

2.2. Searching the databases

The second step involved executing search queries in the chosen databases, which were IEEE, Wiley, Web of Science, Sage, ProQuest, Springer, and Science Direct. The databases were chosen based on their contributions to language education (e.g., Wiley) and AI (e.g., IEEE). Studies were selected by searching for suitable terms—"AI," "artificial intelligence," "EFL," "ESL," "motivation," "attitude," "learning engagement," "learning anxiety," "affective factors" and "higher education"—in the articles' titles and abstracts. To broaden the initial search, no limitations were set on the year of publication. A total of 64 publications were retrieved.

Table 1
Reviewed articles.

No.	Scope of the study	Authors and Year	Purpose of the study	Research deign	Participants	Integrated technology	Task design	Period
1	Motivation	Al-Abdullatif et al., 2023	To investigate the effect of Bashayer, a chatbot embedded into WhatsApp, to support postgraduate students' motivation and learning strategies.	- A quasi-experimental study - Quantitative design using MSLQ	60 Saudi postgraduates	Chatbot	Task-oriented chatbot using text conversation	One Semester
2	Motivation	Ali et al., 2023	To investigate the way ChatGPT impacts learning EFL.	A quantitative research design using an online questionnaire.	80 teachers and EFL students	ChatGPT	Using ChatGPT to enhance language proficiency.	Not specified
3	Motivation	Annamalai et al., 2023	To explore the way Chatbot is used to motivate learners to learn EFL.	A qualitative study using Interviews	25 EFL undergraduate students	The participants were given a choice to use one of the Chatbot application, either Duolingo, Mondly, or Andy	Learning EFL through chatbot	One semester, 3 months
4	Motivation	Moybeka et al., 2023.	To investigate the effectiveness of AI platforms to enhance intrinsic motivation and self-efficacy and facilitate personalized EFL learning experiences.	A quantitative study Using a survey	25 EFL students	ChatGPT	Learning assistant	Not specified
5	Motivation	Kim, 2018	To investigate whether an AI chatbot influence EFL learning.	A quasi-experimental. A qualitative design using Pre-and post-tests	47 Korean EFL learners	Chatbot	EFL Learning assessment	Eight weeks
6	Moivation	Yin et al., 2021	To explore the effect of a chatbot-based micro-learning system on students' learning performance and motivation.	-A quasi-experimental study. A mixed approach using pre-and post-test and questionnaire.	99 Chinese students	Chatbot	The chatbot contains ten micro-learning lessons.	10 micro-learning lessons, each lasting from 8 to 10 min.
7	Learning engagement	Ayedoun et al., 2020	To enhance EFL learners to communicate.	An experimental study. A quantitative method using A questionnaire	60 EFL learners in a Japanese University	Conversational agent	Meaning negotiation	Two-phase experiments
8	Learning engagement	Xie et al., 2021	To design and implement a motivational online conversational agent (MOCA) to enhance learner engagement. To examine the effects of MOCA on promoting positive changes in learning engagement.	A quantitative method using a questionnaire	40 EFL learners from China	Motivational online conversational agent (MOCA)	Multi-turn conversation	Not specified
9	Motivation and learning engagement	Ayedoun et al., 2019	To enhance learners' EFL communication.	An experimental study A quantitative design employed a survey.	40 EFL learners in a Japanese university	Conversational agent	Dialogue interaction and meaning negotiation	One day experiment
10	Motivation and learning engagement	Huang et al., 2023.	To investigate the effects of an artificial Intelligence–agent on learners' engagement, and motivation in a flipped classroom.	A quasi-experimental study using a quantitative method.	102 Taiwan EFL learners, 43 of them in the experimental group	AI agent	Personalized recommendation on learning performance	3 months
11	Attitude	Alhaisoni & Alhaysony, 2017.	To examine EFL learners' attitudes toward GT.	Quantitative using questionnaire	92 Saudi EFL learners	Google translation	Used for educational purposes	Not specified

(*continued on next page*)

Table 1 (*continued*)

No.	Scope of the study	Authors and Year	Purpose of the study	Research deign	Participants	Integrated technology	Task design	Period
12	Attitude	Aljohani, 2021.	To investigate Saudi EFL teachers and students' perceptions of using artificial intelligence (AI) to improve English language learning.	A quantitative study, the researcher employed a close-ended questionnaire.	5 teachers and 16 female students from a Saudi University.	No	No	No
13	Attitude	Arini et al., 2022	To examine the improvement of EFL graduate students in using artificial intelligence-based mobile learning.	A quasi-experimental study A qualitative design using pre-and post-tests.	200 EFL learners, 100 in each group.	Novo Learning's a mobile learning platform powered by AI	Enhances EFL learning in the areas of vocabulary, grammar, listening, pronunciation, and role-play.	One semester
14	Attitude	Chen et al., 2023	To investigate the efficacy of a pedagogical chatbot.	Experimental study through the use of chatbot-guided Interview	215 EFL learners	Chatbot	Conversational agent	One experiment
15	Attitude	Haryanto, & Ali, 2019	To examine the learners' attitudes and perception on using of AI-based SIRI in learning EFL.	A mixed method using a questionnaire and focus group discussion.	40 Indonesian EFL learners	SIRI	learning assistant	One semester
16	Attitude and motivation	Ebadi & Amini, 2022	Use social presence and human-likeness scales to examine EFL learners' attitudes and motivation toward using a task-based Chatbot.	A mixed method using audio-recordings, a questionnaire and semi-structured focus group interview.	256 EFL students	Chatbot	Computer Simulation in Educational Communication (CSIEC) Chatbot	Not specified
17	Attitude and motivation	Lee et al., 2022.	To investigate the impact of an AI-based chatbot on academic performance, self-efficacy, attitude, and motivation.	A quasi-experimental study. A quantitative design using a questionnaire	18 ESP learners	Chatbot	Reviews class material	Two experiments each lasted for 50 min
18	Attitude and learning anxiety	Li & Peng, 2021	To examine a blended teaching mode on an AI language learning platform.	A mixed study based on a quasi-experimental approach, using attitude questionnaire and post-test.	59 EFL learners from China	AI language learning platform	To practice conversation	One semester
19	Learning anxiety	Bao, 2019	To investigate the way an AI-based chatbot is able to reduce anxiety amongst EFL learners.	A mixed method using a questionnaire, face-to-face interviews and results of IELTS speaking part.	40 EFL learners	Chatbot freely available by Google	No	Four weeks
20	Learning anxiety	Hsu et al., 2021	To offer an interactive chatbot system, TPBOT, to eliminate learners' anxiety when speaking EFL.	A quasi-experimental study. A qualitative design using TOEIC speaking scores.	48 Taiwanese EFL learners, 24 of them were assigned to the experimental group.	TPBOT Chatot implemented on LINE platform (an instant messaging platform).	Interactive chatbot to practice spoken English.	Four months
21	Learning anxiety	El Shazly, 2021	To investigate the role of AI on Foreign language anxiety.	A quasi-experimental. A mixed design using pre–and post- speaking test and a questionnaire.	48 Egyptian EFL learners	Chatbot	Interactive spoken-activity	Eight weeks

2.3. Selecting the final sample

The third step involved selecting the final sample for analysis. The search was narrowed to include articles written in English. Additionally, the titles and abstracts of the articles were scrutinized for their relevance to AI applications used to enhance EFL learners' affective factors in higher education. This resulted in the inclusion of 21 studies in the final analysis see Table 1.

2.4. Evaluating the sample

To evaluate the chosen articles, the author developed criteria, and the analysis was carried out using Microsoft Excel. During the process, the studies were categorized into four domains: motivation, attitude, learning engagement, and learning anxiety. Some studies fell into more than one domain, such as motivation and learning engagement or motivation and attitude. In addition, the studies were organized into meaningful subcategories based on study purpose, the examined AI tools and applicability in higher education to learn EFL. Furthermore, the authors reviewed the methodology employed in each study and summarized the findings of the selected articles to develop an adequate analytical framework. In addition, Table 1 includes the AI technologies utilized in EFL education to enhance affective factors. Hence, two dimensions were identified: integrated technology and task design see Table 1.

3. Results

3.1. Motivation and engagement

The use of chatbots in higher education fostered learners' motivation, eagerness, and confidence ([25–27,31,41,42]); increased their intrinsic and extrinsic motivation in language learning; and encouraged them to invest time and effort ([42,43]). More specifically, the experimental group in a study conducted by Yin et al. (2019) was more competent in learning and attained significantly higher intrinsic motivation. Moreover, AI-based conversational agents enhanced learners' motivation to communicate in the target language [26]. In addition, learners were enthusiastically involved in the learning process and actively motivated to use SIRI to learn EFL [44]. Furthermore, conversational agents such as MOCA ([26,45]) enhanced learner engagement in EFL collaborative tasks and significantly improved their learning performance [26] but needed a moderate level of motivation [45].

3.2. Attitude toward using AI-based systems

In general, attitude refers to a positive or negative evaluation of people, things, situations, ideas, or the entire environment. Attitude can help to predict his or her acceptance of novel technologies and applications [11]. Attitude was examined in seven studies. In most studies, learners reported a positive attitude toward learning EFL with the help of AI. For example, Aljohani (2021), Arini et al. (2022), and Li and Peng (2021) are three studies that reported that learners had a positive attitude toward AI-integrated EFL learning and perceived it as an opportunity to improve their English competence.

Additionally, Alhaisoni and Alhaysony (2017) found that learners had a positive attitude toward using a grammar translation AI tool for improving their language proficiency. Another study conducted by Haryanto and Ali (2019) revealed that EFL learners were positive about and willing to use SIRI in EFL learning. Chen et al. (2023) and Ebadi (2022) used chatbots as conversational agents to enhance the EFL learning experience; the results revealed that learners demonstrated a positive attitude toward using chatbots to improve their English language skills and remain motivated. Finally, Li and Peng (2021) examined the impact of a blended teaching mode based on an AI language-learning platform for EFL learners and found that learners had positive attitudes toward using this AI platform, and it helped them reduce their learning anxiety; however, there was no significant difference in language performance between the experimental and control groups see Table 2.

3.3. Learning anxiety

The integration of chatbots to eliminate learning anxiety has shown significant promise in most studies ([28,46,47]). Nonetheless, a study that investigated the role of chatbots in enabling interactive activity to reduce EFL learning anxiety resulted in a significant improvement in linguistic gains, but did not reduce learners' anxiety [48] see Table 2.

4. Discussion

4.1. The role of artificial intelligence tools integrated into EFL classes

Studies have revealed that the most commonly used AI technologies for enhancing learners' affective factors are chatbots and language-learning platforms powered by AI. The AI tools performed three roles in EFL classes in higher education: as teaching assistants, personal tutors, and learning partners. The chatbots that operated as teaching assistants provided academic knowledge ([41,48–50]), offered formative feedback to EFL learners [45], and enabled scaffolded learning ([22,27,31,42,51]). Chatbots also acted as learning partners by chatting and interacting with learners through voice ([26,28,52,47,53]) and text ([27,41]). The role of the chatbot as a personal tutor was to provide valuable language learning contexts; applications such as Duolingo, Mondly, Andy, and SIRI were used as personal tutors ([25,43,44]). [54] and Bao (2019) investigated the integration of AI-based chatbots into EFL classes and found

Table 2
Results.

No.	Authors and Year	Positive or Negative	Statistical analysis
1	Al-Abdullatif et al., 2023	Positive	The mean response for: 1- The task value in the experimental group (4.08) and control group (3.38). 2- The self-efficacy for learning and performance in the experimental group (3.99) and the control group (3.37). 3- Motivation scale in total for the experimental group (M = 4.03) and the control group (M = 3.34).
2	Ali et al., 2023	Positively influenced learning motivation. There is a neutral attitude toward the impact of ChatGPT.	ChatGPT motivates EFL learners to develop: Reading skill (mean 4.02, SD 1.28) Writing skill (mean 3.92, SD 1.016). Grammar (mean 3.9 and SD. 0.9648) Vocabulary (mean of 4.45 and SD 0.6731) There is a neutral attitude towards the impact of ChatGPT on developing listening and speaking skills (M. 3.32, SD 1.28 and M. 3.25, SD 1.28) respectively.
3	Annamalai et al., 2023	Positive	–
4	Moybeka et al., 2023.	Positive	The student learning motivation was at a high level, with a mean score of 4.60.
5	Kim, 2018	Positive	There were significant mean differences between the experimental and control groups with regard to motivation (t = 5.216, p = 0.000), interest (t = 7.882, p = 0.000), and confidence (t = 3.616, p = 0.001)
6	Yin et al., 2021	Positive	All dimensions of intrinsic motivation value correlation coefficients with $p < .01$.
7	Ayedoun et al., 2020	Positive	There was a significant difference in: Confidence F(2,24) = 3.48, p < 0.05 desire F(2,24) = 4.97, p < 0.05 anxiety F(2,24) = 2.88, p < 0.1 All of these resulted in learning engagement.
8	Xie et al., 2021	Positive	The use of MOCA significantly improved learners' engagement scores (p < .001)
9	Ayedoun et al., 2019	Positive	There was a significant difference among the three groups in terms of: confidence F(2, 36) = 7.139, p < 0.05 anxiety F(2, 36) = 3.472, p < 0.1 desire to communicate F(2, 36) = 6.466, p < 0.05 All of these resulted in high motivation and learning engagement.
10	Huang et al., 2023.	Positive Significantly improve the learning engagement with a moderate level of motivation.	–
11	Alhaisoni & Alhaysony, 2017.	Positive	–
12	Aljohani, 2021.	Positive	–
13	Arini et al., 2022	Positive	The t value is (6.373)
14	Chen et al., 2023	Positive	–
15	Haryanto, & Ali, 2019	Positive	–
16	Ebadi & Amini 2022	Positive	–
17	Lee et al., 2022.	Positive	–
18	Li & Peng, 2021	Although there is no statistical difference between the two groups' scores. The learners showed a positive attitude toward the AI language learning platform.	–
19	Bao, 2019	Positive	The experimental group registered a cumulative reduction in anxiety (−31)
20	Hsu et al., 2021	Positive	–
21	El Shazly, 2021	Negative	The mean anxiety value in the pre-and post-intervention was 15.3 and 16.8, respectively. The learners became more anxious.

that teachers' and students' responded positively toward the integration, and students experienced reduction in learning anxiety; however, these studies did not specify the main task of the AI in the EFL class.

Given the functions of AI tools, it is most likely that these technologies will significantly and positively influence language learning achievement, engagement, motivation, and attitudes. In addition, the introduction of chatbot systems in the educational sector has the potential to provide personalized EFL learning experiences, by adapting to learners' needs and providing learning materials anywhere and anytime. These chatbots performed different tasks, such as meaning negotiation, interactive conversation, learning assistance, storytelling, and answering questions. The use of such chatbots resulted in better academic achievement, enhanced motivation and engagement, reduced anxiety, and increased attention and collaboration.

4.2. Improvements in motivation, engagement, and attitude

4.2.1. Learning motivation and engagement

Motivation is the learner's choice to participate in a specific learning activity and the energy devoted to sustain throughout the learning journey [55]. Therefore, in the context of language learning, whereas motivation acts as a driving force toward specific behaviors, engagement concerns with the learner's active involvement in learning tasks [56].

Interactive tools such as chatbots kept learners motivated and engaged, while allowing them to learn EFL in an exciting and relaxing atmosphere ([26,41,42,45]). Learners were not bored while learning with a conversational agent, and could quickly and easily acquire knowledge. Furthermore, educators have realized the significance of utilizing AI technologies in educational contexts to provide learners with engaging and interactive experiences ([27,43,53]). In addition, voice-based chatbots positively influenced EFL learners' motivation ([27,52]). Moreover, chatbot-assisted microlearning systems effectively promoted intrinsic motivation [43].

Motivation identifies the learners' intent, whereas engagement emphasizes their actions. Four of the reviewed studies examined the effects of AI agents on EFL learner engagement ([26,45,46]). These studies revealed that AI-based conversational chatbots significantly improved EFL learners' engagement.

The use of AI technology has shown positive effects on EFL learners' motivation and engagement. These technologies provide opportunities to boost intrinsic and extrinsic motivation; enhance learners' confidence, eagerness, and self-efficacy; and foster their engagement.

4.2.2. Learners' attitude

Throughout the review, it has been highlighted that learners have a positive attitude toward the application of AI in EFL learning. AI-based learning platforms, chatbots, and AI agents have created interactive, responsive, and motivating environments and increased opportunities for EFL learning, thereby enhancing learners' positive attitudes toward language learning.

4.3. Reduction in learning anxiety

In language learning, chatbots also have the potential to reduce learning anxiety ([28,46,47,49]). However, in a study conducted by El Shazly (2021), interactions with a chatbot did not reduce learners' anxiety; it only resulted in a significant improvement in linguistic gains.

It is evident that these AI technologies have the potential to positively impact various aspects of EFL learning, such as improving motivation, engagement, and attitudes and reducing anxiety in EFL higher education. In addition to enhancing learners' affective factors, the use of AI tools has led to improvements in EFL learning performance. In all the reviewed studies, the experimental groups using chatbots and AI agents had better learning outcomes than the control groups. This indicates that AI technologies are effective in fostering the acquisition of knowledge and the development of language skills.

Despite the promising outcomes of AI technologies, the studies reviewed emphasize the significance of a teacher's role in the EFL learning process. Teachers play a key role in facilitating the use of AI technology, providing guidance, and fostering motivation. Therefore, AI must be considered a complementary tool rather than a substitute for a teacher.

5. Conclusion and practical implications

This study presents a systematic review of papers published between 2017 and 2023 on the use of AI applications in higher education to support EFL learners' affective factors. These affective factors were classified as motivation, engagement, attitude, and learning anxiety. Generally, the information from the reviewed articles emphasized that there is a promising future for AI technologies in EFL education. AI has had a positive impact by optimizing English language skills and learners' motivation, engagement, and attitudes. This is evident from the increase in the number of recently published articles. It is evident that Japanese and Chinese scholars account for 50 % of published articles in the fields of AI and EFL learning [55].

Nevertheless, AI integration into EFL education is in its early stage, and further research is necessary. There is a scarcity of research on integrating AI to enhance EFL learners' affective factors. Most studies have focused on employing conversational chatbots (to strengthen listening and speaking skills) and automated writing-evaluation systems (to improve writing skills and to correct vocabulary and grammar usage), ignoring the significance of affective factors that might negatively or positively affect the overall learning performance. In a systematic review, Zawacki-Richter et al. (2019) noted, "More research is needed from educators and learning designers on how to integrate AI applications throughout the student lifecycle" (p. 22), which supports the findings of the current study.

Furthermore, there has been insufficient elaboration on the correct instructional method to implement when integrating AI technologies into EFL classrooms. Moreover, further research is necessary to shed new light on pedagogical and ethical implications. Additionally, the lack of longitudinal studies, the pervasiveness of quantitative methods, and the prevalence of quasi-experimental designs emphasize that there is still an abundance of opportunities for educators to pursue novel and valuable research using AI technologies in different fields of higher education and, more specifically, English language education [35].

The findings in this emerging area of research could be significant for learners, researchers, educators, and language-program designers. Considering EFL learners' positive attitude toward AI-integrated EFL classes, it is reasonable to suppose that learners will embrace the use of AI applications in learning EFL. The current study also helps researchers better grasp how the literature is progressing and identify the areas of AI currently being studied. Educators can even rely on the results as a guide to develop their

teaching strategies and evaluation procedures and to design AI-based learning activities. Finally, language-program designers can use the findings of this study to evaluate the efficiency of AI applications and employ them effectively in developing EFL curricula. The currently available AI applications provide pedagogical opportunities to design intelligent learning systems that scaffold learning; provide adaptive learning experiences; and enable interactive and personalized learning opportunities to foster motivation, engagement, and positive attitudes [17].

However, the use of AI in education is a cause of grave concern worldwide, especially after UNESCO's 2019 consensus. UNESCO published a document titled "Beijing Consensus on Artificial Intelligence (AI) and Education" to offer guidance on implementing AI technologies to achieve the objectives of the 2030 Education Agenda. Therefore, it is important to be concerned about other AI dimensions such as the ethical, cultural, social, and economic dimensions, to ensure inclusion and equity, prepare educators for AI-based education, develop quality and inclusive learning systems, and verify ethics and transparency [57].

Limitation and recommendations

The papers reviewed supported the significance of AI technology in enhancing EFL learners' affective factors. However, various gaps need to be addressed in future research. Most previous studies do not specify the teaching approach implemented in the learning process. A complete description of the teaching method and design should be provided so that stakeholders can fully benefit from best practices and understand the conditions under which AI is used to optimize learning [55].

Some of the articles reviewed in this study needed to clearly present whether the researchers examined the validity and reliability of the data collection methods, which included questionnaires, pre- and post-tests, and integrated AI technologies. Moreover, only a few researchers conducted pilot studies before implementing AI technology in the EFL learning context. The absence of such details might have affected the results and is a major flaw affecting the accuracy and feasibility of the study [58].

Studies have actively delineated productive and receptive skills. In contrast, affective factors, such as motivation and attitude toward language learning, which facilitate or hinder the learning process, are covered the least in recent studies. This paucity may indicate the need for further research to investigate the adaptability of AI: adaptability to EFL learners with varying affective factors (such as language aptitude, motivation, learning engagement, attitude, and anxiety) and adaptability to different learning and teaching contexts, such as blended teaching [59].

The most employed data collection methods were quantitative, such as questionnaires. Future studies can use other qualitative methods to provide a deeper understanding of students' experiences with AI technologies, and this can enable the researcher to present the results in greater detail than with just quantitative studies. The mixed data collection method was employed in six studies and included questionnaires, pre-and post-tests, semi-structured interviews, and audio recordings.

In terms of the AI technology used to enhance EFL learners' affective factors and reduce their anxiety in language learning, AI agents such as SIRI, ChatGPT, and chatbots implemented in learning platforms were only used as an Intelligent tutors and assistants. Other AI tools could be integrated, to guide and support learners in performing language learning tasks effectively and to diagnose their learning problems. Therefore, diverse applications of AI in language learning must be integrated to offer personalized learning experiences, and the user interfaces should be customized according to learners' learning states to foster motivation, engagement, and positive attitudes.

In addition, there is a scarcity of longitudinal studies investigating the effectiveness of AI technologies in EFL contexts through experiments with large groups of participants, validated methods, competent teachers, and supporting educational institutions. Most experiments were conducted over a brief period, ranging from two 50-min experiments [50] to a three-month quasi-experimental study ([41,45,47,51]).

Most of the reviewed studies indicated that AI applications can motivate students and foster their learning engagement. However, these benefits have occurred for competent learners, as AI technology has not yet been well developed for integration into the learning process. In addition, teachers should be more knowledgeable about the appropriate application of AI. Students may become demotivated by the use of inappropriate learning resources or if they find using AI technologies challenging. Therefore, teachers must receive sufficient training to effectively understand and utilize AI applications. Going forward, learners need to know about appropriate AI applications in learning, and adequacy of their practice should be considered [60].

Data availability

Due to the nature of the systematic review, data availability is not applicable.

CRediT authorship contribution statement

Lujain AlTwijri: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Talal Musaed Alghizzi:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Y.J. Fandiño Parra, Explicit teaching of socio-affective language learning strategies to beginner EFL students, Íkala, Rev. Leng. Cult. 15 (24) (2010) 145–169.
[2] J. Arnold, Affective factors and language learning, Language education and emotions: Research into emotions and language learners, language teachers and educational processes (2020) 18–33.
[3] R. Nishant, M. Kennedy, J. Corbett, Artificial intelligence for sustainability: challenges, opportunities, and a research Agenda, Int. J. Inf. Manag. 53 (2020) 102104.
[4] E.R. Lai, Motivation: a literature review, Person Research's Report 6 (2011) 40–41.
[5] J.E. Ormrod, How motivation affects learning and behavior, Educ. Week (2010). Obtained from, http://www.education.com/reference/article/motivation-affects-learning-behavior.
[6] S.M. Gass, J. Behney, L. Plonsky, Second Language Acquisition: an Introductory Course, fifth ed., Routledge, 2020 https://doi.org/10.4324/9781315181752.
[7] L. Ortega, Understanding Second Language Acquisition, Routledge, 2014.
[8] R.M. Ryan, E.L. Deci, *Self-determination theory*, Basic Psychological Needs in Motivation, Development, and Wellness (2017).
[9] K.A. Noels, L.G. Pelletier, R. Clément, R.J. Vallerand, Why are you learning a second language? Motivational orientations and self-determination theory, Lang. Learn. 50 (1) (2000) 57–85.
[10] P. Hiver, A.H. Al-Hoorie, J.P. Vitta, J. Wu, Engagement in language learning: a systematic review of 20 Years of research methods and definitions, Lang. Teach. Res. 28 (1) (2024) 201–230.
[11] H.D. Brown, Principles of Language Learning and Teaching, vol. 4, Longman, New York, 2000.
[12] R. Ellis, The Study of Second Language Acquisition, Oxford University Press, Oxford, UK, 1994, p. 824, vii.
[13] T.R.A. Mukhallafi, Using artificial intelligence for developing English language teaching/learning: an analytical study from university students' perspective, Int. J. Engl. Ling. 10 (6) (2020) 40.
[14] L. Han, Analysis of new advances in the application of artificial intelligence to education, in: 2018 3rd International Conference on Education, E-Learning and Management Technology (EEMT 2018), Atlantis Press, 2018, pp. 608–611.
[15] M.L. Owoc, A. Sawicka, P. Weichbroth, Artificial intelligence technologies in education: benefits, challenges and strategies of implementation, in: IFIP International Workshop on Artificial Intelligence for Knowledge Management, Springer International Publishing, Cham, 2019, August, pp. 37–58.
[16] L. Chen, P. Chen, Z. Lin, Artificial intelligence in education: a review, IEEE Access 8 (2020) 75264–75278.
[17] M. Fahimirad, S.S. Kotamjani, A review on application of artificial intelligence in teaching and learning in educational contexts, Int. J. Learn Dev. 8 (4) (2018) 106–118.
[18] M. Haenlein, A. Kaplan, A brief history of artificial intelligence: on the past, p present, and future of artificial intelligence, Calif. Manag. Rev. 61 (4) (2019) 5–14.
[19] C. Zhai, A systematic review on artificial intelligence dialogue systems for enhancing English as foreign language students' interactional competence in the university, Comput. Educ.: Artif. Intell. (2023) 100134.
[20] T. Chu, W. Wang, X. Chen, Visual analysis of research trends on language teachers in the era of artificial intelligence, Advances in Educational Technology and Psychology 7 (8) (2023) 71–76.
[21] N. Haristiani, Artificial intelligence (AI) chatbot as language learning medium: an inquiry, J. Phys. Conf. 1387 (1) (2019, November) 012020. IOP Publishing.
[22] N.Y. Kim, Chatbots and Korean EFL students' English vocabulary learning, Journal of Digital Convergence 16 (2) (2018).
[23] H.S. Kim, Y. Cha, N.Y. Kim, Effects of AI Chatbots on EFL Students' Communication Skills, vol. 21, 2021, pp. 712–734.
[24] R. Alsadoon, Chatting with AI bot: vocabulary learning assistant for Saudi EFL learners, Engl. Lang. Teach. 14 (6) (2021) 135–157.
[25] N. Annamalai, M.E. Eltahir, S.H. Zyoud, D. Soundrarajan, B. Zakarneh, N.R. Al Salhi, Exploring English language learning via chabot: a case study from a self determination theory perspective, Comput. Educ.: Artif. Intell. (2023) 100148.
[26] E. Ayedoun, Y. Hayashi, K. Seta, Adding communicative and affective strategies to an embodied conversational agent to enhance second language learners' willingness to communicate, Int. J. Artif. Intell. Educ. 29 (1) (2019) 29–57, https://doi.org/10.1007/s40593-018-0171-6.
[27] S. Ebadi, A. Amini, Examining the roles of social presence and human- likeness on Iranian EFL learners' motivation using artificial intelligence technology: a case of csiec chatbot, Interact. Learn. Environ. (2022) 1–19.
[28] Chen Hsu, C.-S. Yu, Proposing a task-oriented chatbot system for EFL learners speaking practice, Interact. Learn. Environ. (2021) 1–12.
[29] Y. Turan, Role of affective factors in foreign language learning, Journal of Educational & Instructional Studies in the World 8 (3) (2018).
[30] I.P. Hapsari, T.T. Wu, AI chatbots learning model in English speaking skill: alleviating speaking anxiety, boosting enjoyment, and fostering critical thinking, in: International Conference on Innovative Technologies and Learning, Springer International Publishing, Cham, 2022, pp. 444–453.
[31] A.M. Moybeka, N. Syariatin, D.P. Tatipang, D.A. Mushthoza, N.P.J.L. Dewi, S. Tineh, Artificial intelligence and English classroom: the implications of AI toward EFL students' motivation, Edumaspul: Jurnal Pendidikan 7 (2) (2023) 2444–2454.
[32] T.K. Chiu, Q. Xia, X. Zhou, C.S. Chai, M. Cheng, Systematic literature review on opportunities, challenges, and future research recommendations of artificial intelligence in education, Comput. Educ.: Artif. Intell. 4 (2023) 100118.
[33] A. Nigam, R. Pasricha, T. Singh, P. Churi, A systematic review on AI-based proctoring systems: past, present and future, Educ. Inf. Technol. 26 (5) (2021) 6421–6445.
[34] F. Tahiru, AI in education: a systematic literature review, J. Cases Inf. Technol. 23 (1) (2021) 1–20.
[35] O. Zawacki-Richter, V.I. Marín, M. Bond, F. Gouverneur, Systematic review of research on artificial intelligence applications in higher education–where are the educators? International Journal of Educational Technology in Higher Education 16 (1) (2019) 1–27.
[36] C. Zhai, S. Wibowo, A WGAN-based DialoColgue system for embedding humor, empathy, and cultural aspects in education, IEEE Access (11) (2023) 71940–71952.
[37] F. Ouyang, L. Zheng, P. Jiao, Artificial intelligence in online higher education: a systematic review of empirical research from 2011 to 2020, Educ. Inf. Technol. 27 (6) (2022) 7893–7925.
[38] H.L. Colquhoun, D. Levac, K.K. O'Brien, S. Straus, A.C. Tricco, L. Perrier, et al., Scoping reviews: time for clarity in definition, methods, and reporting, J. Clin. Epidemiol. 67 (12) (2014) 1291–1294.
[39] J.F. Wolfswinkel, E. Furtmueller, C.P. Wilderom, Using grounded theory as a method for rigorously reviewing literature, Eur. J. Inf. Syst. 22 (1) (2013) 45–55.
[40] J. Firmstone, Editorial Journalism and Newspapers' Editorial Opinions, Oxford research encyclopedia of communication, 2019, https://doi.org/10.1093/acrefore/9780190228613.013.803.
[41] A.M. Al-Abdullatif, A.A. Al-Dokhny, A.M. Drwish, Implementing the bashayer chatbot in Saudi higher education: measuring the influence on students' motivation and learning strategies, Front. Psychol. 14 (2023) 1129070.
[42] E. Ayedoun, Y. Hayashi, K. Seta, Toward personalized scaffolding and fading of motivational support in L2 learner–dialogue agent interactions: an exploratory study, IEEE Transactions on learning technologies 13 (3) (2020) 604–616, https://doi.org/10.1109/TLT.2020.2989776.
[43] J.K.M. Ali, M.A.A. Shamsan, T.A. Hezam, A.A. Mohammed, Impact of ChatGPT on learning motivation: teachers and students' voices, Journal of English Studies in Arabia Felix 2 (1) (2023) 41–49, https://doi.org/10.5539/elt.v14n6p135. Learners. English Language Teaching, 14(6), 135–157.
[44] E. Haryanto, R.M. Ali, Students' attitudes towards the use of artificial intelligence SIRI in EFL learning at one public university, BKS-PTN Wilayah Barat 1 (1) (2019).
[45] T. Xie, R. Liu, Y. Chen, G. Liu, MOCA: a motivational online conversational agent for improving student engagement in collaborative learning, IEEE Transactions on Learning Technologies 14 (5) (2021) 653–664.
[46] A.Y. Huang, O.H. Lu, S.J. Yang, Effects of artificial intelligence–enabled personalized recommendations on learners' learning engagement, motivation, and outcomes in a flipped classroom, Comput. Educ. 194 (2023) 104684.
[47] M. Bao, Can home use of speech-enabled artificial intelligence mitigate foreign language anxiety: investigation of a concept, Arab World Engl. J. 5 (2019) 28–40.
[48] R. El Shazly, Effects of artificial intelligence on English speaking anxiety and speaking performance: a case study, Expet Syst. 38 (3) (2021) e12667.

[49] B. Li, M. Peng, The evaluation of a blended teaching mode based on an AI language learning platform, in: The Second International Conference on Information Science and Education, 2021, pp. 1017–1021, https://doi.org/10.1109/ICISEIE53922.2021.00320.
[50] E. Alhaisoni, M. Alhaysony, An investigation of Saudi EFL university students' attitudes towards the use of google translate, Int. J. Engl. Lang. Educ. 5 (1) (2017) 72–82.
[51] Y. Lee, G. Hwang, P. Chen, Impacts of an AI-based chabot on College students' after-class review, academic performance, self-efficacy, learning attitude, and motivation, Educ. Technol. Res. Dev. 70 (5) (2022) 1843–1865.
[52] J. Yin, T.T. Goh, B. Yang, Y. Xiaobin, Conversation technology with micro- learning: the impact of chatbot-based learning on students' learning motivation and performance, J. Educ. Comput. Res. 59 (1) (2021) 154–177.
[53] D.N. Arini, F. Hidayat, A. Winarti, E. Rosalina, Artificial Intelligence (AI)- Based Mobile Learning in ELT for EFL Learners: the Implementation and Learners, Attitudes, 2022.
[54] R.A. Aljohani, Teachers and students' perceptions on the impact of artificial intelligence on English language learning in Saudi arabia, Journal of Applied Linguistics and Language Research 8 (1) (2021) 36–47.
[55] L. Koff, R. Mullis, Nutrition education and technology: can delivering messages via new media technology effectively modify nutrition behaviors of pre-schoolers and their families? J. Nutr. Educ. Behav. 43 (4) (2011) S40. S40.
[56] Y. Chen, S. Jensen, L.J. Albert, S. Gupta, T. Lee, Artificial intelligence (AI) student assistants in the classroom: designing chatbots to support student success, Inf. Syst. Front 25 (1) (2023) 161–182.
[57] T.A. Sharadgah, R.A. Sa'di, A systematic review of research on the use of artificial intelligence in English language teaching and learning (2015-2021): what are the current effects? J. Inf. Technol. Educ. 21 (2022).
[58] J.W. Creswell, J.D. Creswell, Research Design: Qualitative, Quantitative and Mixed Methods Approaches, fifth ed., SAGE, Los Angeles, 2018.
[59] F. Pedro, M. Subosa, A. Rivas, P. Valverde, Artificial Intelligence in Education: Challenges and Opportunities for Sustainable Development, 2019.
[60] R. Jiang, How does artificial intelligence empower EFL teaching and learning nowadays? A review on artificial intelligence in the EFL context, Front. Psychol. 13 (2022) 1049401.

The Corresponding author is Lujain AlTwijri

Lujain Abdulrahman AlTwijri is a PhD candidate in Applied Linguistics at Imam Mohammad Ibn Saud Islamic University, Riyadh, Saudi Arabia. She has obtained her BA in English Language and Literature from Princess Nora bint Abdulrahman University and her MA in Linguistics from Imam Mohammad Ibn Saud Islamic University. She has a TESOL certificate from Montclair State University, New Jersey, USA.

ORCID account: https://orcid.org/0009-0000-0590-5034

E-mail: altwijri-lujain@hotmail.com

Dr. Talal Musaed Alghizzi is an associate professor in Applied

Linguistics at the Department of English Language and Literature, Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh, Saudi Arabia. He has obtained his BA in English Language and Literature and an MA in Translation from IMSIU. Also, Dr. Alghizzi has obtained a second MA as well as a PhD in Applied Linguistics from University College Cork, Cork, Ireland. Dr.

Alghizzi's research interests are: translation, language skills, and technology integration.

ORCiD account: http://orcid.org/0000-0002-1088-9216

E-mail: tmalghizzi@imamu.edu.sa